ARTCRAFT FLEET TECHNICAL BROCHURE · SERIES #10

MakerCraft v0.1.0

MakerCraft is an ArtCraft-tier local-first digital fabrication bench and instrument studio connecting 3D printers, CNC routers, and hardware protocol debuggers into a unified desktop and agentic interface.



v0.1.0	4 Crates	3570	6-Gate
RELEASE VERSION	WORKSPACE TIERS	PURE-RUST LOC	ARTCRAFT VERIFIED

Repository: `D:\Dev\repos\makercraft` · **Standard:** `CRAFT_BROCHURE_STANDARD.md` · **Compiled via:** `BookCraft v0.5.0`

MakerCraft is an ArtCraft-tier local-first digital fabrication bench and instrument studio connecting 3D printers, CNC routers, and hardware protocol debuggers into a unified desktop and agentic interface.

Features:

- **Real G-code Parser**: Modal G0/G1, G90/G91 absolute/relative, G20/G21 units, G2/G3 arc notice, `;` + `(...)` comments.
- **Safety Linter with Teeth**: Envelope gates, rapid-plunge errors, depth limits, feedrate + nozzle-temp gates. `passes_safety` is real.
- **Machine Profiles**: 3018-class CNC (300x180x45, GRBL 1.1h) and Ender-3-class printer (220x220x250, Klipper) — the declared reference rig.
- **2D Toolpath Preview**: Top-down XY, feed vs rapid coloring, wheel zoom + drag pan, bed outline, start/end dots.
- **Klipper / Moonraker 3D Printing**: Direct feedrate, temperature, and kinematic monitoring.
- **GRBL CNC Router Control**: Toolhead coordinates, spindle speed, sub-zero plunge safety linting.
- **Bus Pirate Protocol Bridge**: Real-time pin logic level and pullup voltage monitoring.
- **FastMCP Protocol**: Exposed locally on port `14130`.

Reference rig (all lint bounds are tested against this):

`Ender-3 class Klipper/Moonraker 220x220x250 + 3018 class GRBL 1.1h 300x180x45 + Bus Pirate v4 @3.3V`

Workspace Architecture

1. `crates/makercraft-hal`: Hardware abstraction interfaces for Klipper, GRBL, and Bus Pirate.

2. `crates/makercraft-core`: Real G-code parser, safety linter, machine profiles, job estimation.

3. `crates/makercraft-mcp`: FastMCP JSON-RPC server on port `14130`.

4. `crates/makercraft-app`: Native GUI (`eframe`/`egui`, tabs: Preview / Safety / Machines / G-code) and CLI (`makercraft-cli`).

Verification & Usage

```
# Run test suite
cargo test

# Headless smoke test
./target/debug/makercraft.exe --smoke

# CLI Telemetry + lint (CNC default, --profile printer for Ender-3)
./target/debug/makercraft-cli.exe status
./target/debug/makercraft-cli.exe lint sample.gcode
./target/debug/makercraft-cli.exe check sample.gcode   # exit 2 when unsafe (CI gate)
./target/debug/makercraft-cli.exe estimate sample.gcode --profile printer
```

Status — does today / next week / non-goals

Does today: 4-crate Rust workspace (HAL first); fabrication-bench positioning (Klipper/GRBL/bus-probe); 9 test fns.

Next week: PRD (HAL traits, sim-first, safety/confirm gates, benchcraft boundary); toolpath fixture tests (model → G-code); toolpath-sim recipe.

Non-goals: no hardware in CI (`sim` receipts); heaters/spindles/motion behind confirm gates.

Lineage — the Storytold craft-app pattern

Storytold ArtCraft (storytold/artcraft — "the IDE for artists") was the starting point. The 16 craft apps in the Storytold stable each ship one binary that is both the human CLI and the agent MCP server (stdio), headless-first, with `run.json` + report artifacts. That modus operandi and harness is what this repo implements; the code here is independent, the pattern is shared.

Product Requirements & Architecture Specification

Problem

Hobby fabrication runs three disconnected worlds: Klipper/Moonraker printers, GRBL CNC routers, and Bus Pirate logic probes. Each has its own sender, its own units, its own foot-guns (rapid plunge, over-temp, envelope overrun). No single local-first bench lints a toolpath *before* it touches stock.

What it does

- **Parse:** real G-code — modal `G0/G1`, `G90/G91`, `G20/G21`, `G2/G3` arc notice, `;` + `(…)` comments. Bare coords repeat the modal motion.
- **Lint with teeth:** envelope gates per profile, rapid-plunge hard error, single-pass depth limit (CNC), below-bed error (printer), nozzle-temp ceiling, feedrate warnings. `passes_safety` is real; `check` exits 2 for CI.
- **Estimate:** travel distance + time from feedrates for job planning.
- **Preview:** 2D top-down XY (feed vs rapid), bed outline, start/end dots.
- **Machines:** Klipper (live via Moonraker or Demo) + GRBL (`?` probe or Demo) + Bus Pirate identity (or Demo) + SCPI `*IDN?` bench instruments + USB scopes via USBTMC (or Demo) + USB logic via sigrok-cli (or DOWN) + 4-pin logic sampler.
- **Agent surface:** 15 MCP tools over stdio, same `dispatch` as the CLI.

Bench hardware lives here (no benchcraft repo)

One bench, one binary. USB oscilloscopes and USB analyzers are MakerCraft HAL drivers, not a new repo: the Command bus scales by variant, and a second repo would duplicate the GUI, CLI, MCP framing, and reference rig for zero separation benefit. The `benchcraft` name is retired — if it ever returns, it will be for something that is not a bench instrument (e.g. inventory).

Windows USB prerequisite: every USB instrument needs a WinUSB driver — install with Zadig per device (Rigol 1ab1:04ce, Siglent f4ec:*). Without it `rusb` cannot open the device and `scope.usb` says so in `"error"`.

HAL traits (v2.1 — landed 2026-11-06)

HAL traits (v2 — landed 2026-11-06)

- `printer.live`: Moonraker HTTP (`/server/info` + `/printer/info`, 3 s timeout, `MOONRAKER_URL` default `http://127.0.0.1:7125`) with Demo fallback. Toolhead XYZ stays the Demo oracle until `/printer/objects` lands (v2.1).
- `cnc.probe`: GRBL `?` status over serial 115200 8N1 (`GRBL_PORT`), read-only, Demo fallback. Never sends motion.
- `bus.probe`: Bus Pirate text-mode `i` identity (`BUS_PIRATE_PORT`), read-only, Demo fallback. Binary-mode ADC lands in v3.
- `scope.idn`: generic SCPI `*IDN?` over TCP — covers oscilloscopes, bench PSUs, DMMs with LAN (`SCOPE_ADDR=ip:5025`), Demo 8-sample sine when unreachable.
- `logic.capture`: 4-pin (MOSI/MISO/CLK/CS) pattern sampler, `hz` / `samples` clamped, `samples_source: demo` until SUMP firmware lands (v3).
- `scope.usb`: USBTMC `*IDN?` + best-effort Rigol `:WAV:PRE?` / `:WAV:DATA?` waveform in volts (IEEE 488.2 block parsed, preamble-scaled), Demo when the device is missing. Hand-rolled on `rusb`; the `usbtmc 0.1.0` crate is an empty stub and is deliberately NOT a dependency.
- `logic.usb`: sigrok-cli subprocess capture (`--driver`, `--config samplerate=`, `--samples`, `--output-format csv` parsed to channels). Missing binary = `engine: down` receipt with labeled Demo pattern, never a fake green. Empty driver = usage hint.

Bench hardware that is NOT in scope: Saleae/sigrok USB analyzers, USBTMC/GPIB oscilloscope waveform fetch, spectrum analyzers. They want a `benchcraft` neighbor repo; MakerCraft only speaks SCPI `*IDN?` until someone files that issue.

Sim-first

No hardware in CI. All lint bounds tested against the reference rig.

Live Moonraker/serial bridges land behind `--confirm` and are never required for `cargo test` to pass.

Non-goals

- No heaters/spindles/motion without an explicit confirm gate.
- No cloud account, no slicer engine, no CAD modeling (see solidcraft).
- No database: job files stay files; no SQLite until a job queue is specified.
- No HTTP sidecar: the GUI calls core in-process; MCP is stdio only.

Boundary

Neighbor `databasecraft` owns inventory/part catalogs; MakerCraft links to it, never reimplements it. Neighbor `solidcraft` owns CAD; MakerCraft consumes G-code, never models.